FOROUGH SHIRIN ABKENAR

Ph.D.

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PROFESSIONAL SUMMARY

ML Researcher with expertise in **deep learning**, **generative AI**, **federated learning**, and **domain adaptation**. Experienced in developing **cutting-edge AI models** for **failure prediction**, **optimization**, and **decision-making** in **dynamic environments**. Strong programming skills in **Python** and **PyTorch**, with a deep understanding of **probabilistic modeling**, **representation learning**, and **AI-driven automation**. Passionate about advancing state-of-the-art ML techniques and applying them to real-world challenges.

PROFESSIONAL BACKGROUND

Postdoctoral Scholar

University of California Davis

February 2023 – February 2025 Davis, CA

- Designed **deep learning models** and FL for **failure prediction** and **anomaly detection**, improving predictive accuracy.
- Developed **domain adaptation techniques** (DANN, self-supervised learning) to improve **model generalization** across different datasets.
- Led research on **knowledge distillation**, enhancing model efficiency for deployment in resource-constrained environments.
- Mentored PhD students in **machine learning**, **optimization techniques**, and **deep learning architectures**.

Postdoctoral Scholar

University of California Irvine

February 2022 – February 2023 Irvine, CA

- Developed **adaptive machine learning models** for time-variant datasets, improving performance in **real-world AI applications**.
- Led **feature engineering efforts** and **statistical analysis** for large-scale datasets, optimizing data-driven insights.
- Worked on **reinforcement learning** applications for adaptive decision-making and resource allocation problems.
- Collaborated with **Intel AI/ML teams** on research projects involving **AI-driven automation** and **real-time data processing**.

EDUCATION

Ph.D. in Electrical and Information Engineering

The University of Sydney

Mar 2018 – Jan 2022

Thesis title: Towards Hyper-efficient IoT Networks Using Fog Paradigm

M.Sc. in IT Engineering

Sahand University of Technology

Sept 2012 – June 2014

B.Sc. in IT Engineering

Shahid Madani University of Azarbayjan

Sept 2008 – June 2012

SKILLS

CNN | GAN | VAE | LLM | FL | RL

Python | PyTorch | Scikit-learn | C++ | SQL | MATLAB

Deep Learning | Transfer Learning | Statistical Analysis | Feature Engineering | Optimization | Probabilistic Modeling

ACHIEVEMENTS

- **Generative AI with LLMs Certification**, Amazon Web Services, 2025.
- **Dean's Award for the Best Thesis**, The University of Sydney, 2023.

SELECTED PUBLICATIONS

Forough Shirin Abkenar, et al.:

Full list: GoogleScholar

Journal Articles

- "A Survey on Mobility of Edge Computing Networks in IoT: State-of-the-Art, Architectures, and Challenges," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 4, pp. 2329–2365, 2022.
- "ENERGENT: An Energy-Efficient UAV-Assisted Fog-IoT Framework for Disaster Management," *Journal of Communications and Networks*, vol. 24, no. 6, pp. 698–709, 2022.

Conference Proceedings

- "Online Domain Adaptive Classification for Mobile-to-Edge Computing," in *IEEE WoWMoM*, Boston, MA, 2023.
- "Selective Data Offloading in Edge Computing for Two-Tier Classification With Local Domain Partitions," in *IEEE PerCom Workshops: Cloud2Things*, Atlanta, GA, 2023.